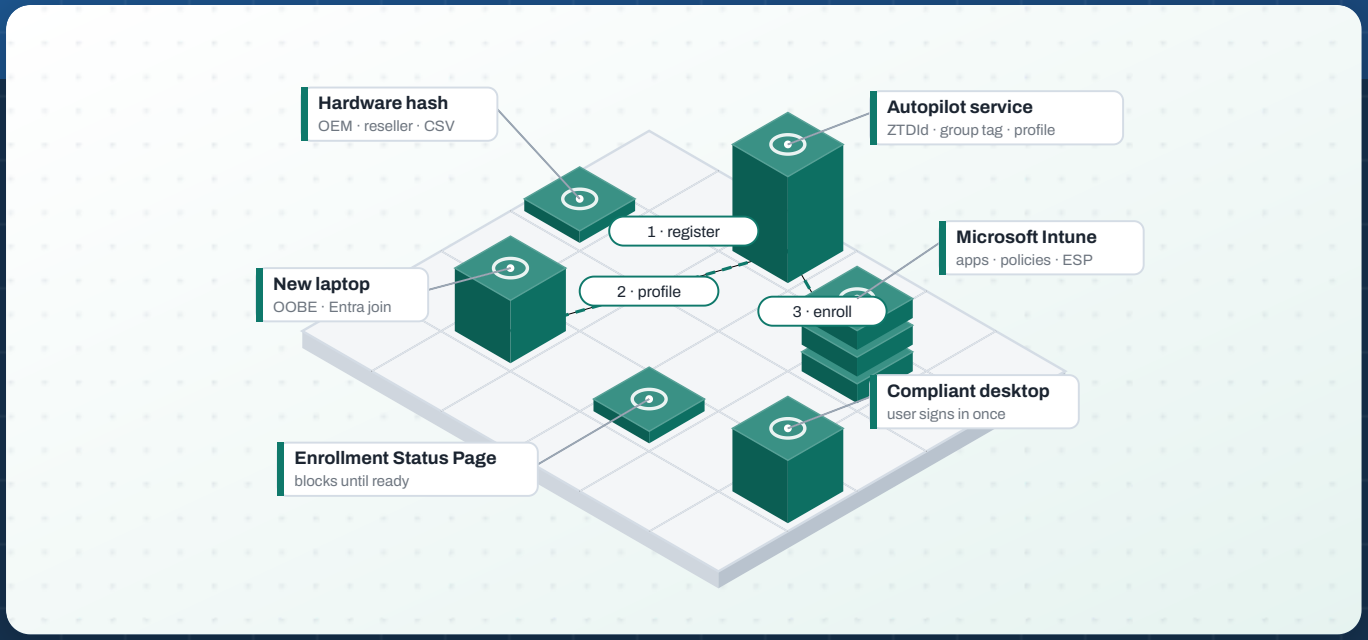
STANDARD OPERATING PROCEDURE · FIELD GUIDE SERIES

SOP-EM-004

Provisioning Windows Devices with Windows Autopilot and Intune

Register the hardware hash, target devices with a ZTDId dynamic group, build the deployment profile and Enrollment Status Page, then hand a sealed laptop to a user who signs in and gets a managed, compliant desktop. Covers user-driven, pre-provisioned and self-deploying modes, and Autopilot device preparation.

Endpoint Management | Windows Autopilot | Microsoft Intune



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LEARNING OBJECTIVES

After working through this procedure you will be able to:

- Explain what the hardware hash, ZTDId and group tag are, and how a device finds its deployment profile during OOBE.
- Choose between user-driven, pre-provisioned and self-deploying modes, and between classic Autopilot and Autopilot device preparation.
- Register devices from the OEM, from a CSV or with Get-WindowsAutopilotInfo, and target them with a dynamic device group.
- Build the deployment profile and Enrollment Status Page so the user reaches a desktop only when it is secure and usable.
- Diagnose a failed deployment from the ESP error, the MDM diagnostics CAB and the Autopilot event logs, and reset or deregister a device safely.

Time	Difficulty	Minimum access	Risk
2–4 h build · 30–60 min per device	Intermediate	Intune Administrator (or Policy and Profile Manager)	Low for new devices; moderate for reset/deregister

CHAPTER 1

Concepts You Need First

Traditional imaging copies a prepared disk onto every PC, then spends hours joining, patching and installing on top of it. **Windows Autopilot** takes the opposite approach: it keeps the Windows installation the manufacturer shipped and *transforms* it into a corporate device the first time it is switched on. The device recognises that it belongs to the organisation, joins Microsoft Entra ID, enrolls in Intune, and pulls down its policies and apps before the user ever reaches the desktop.

KEY CONCEPT · A RESERVATION, NOT AN IMAGE

Think of Autopilot as a reservation at a hotel. The hardware hash is the booking reference the manufacturer files in advance. When the device arrives at the front desk (the out-of-box experience) and gives its reference, the hotel already knows whose it is, which room to give it (the deployment profile) and what has to be in the room before the guest is let in (the Enrollment Status Page). Nothing is copied onto the disk; everything is delivered by policy.

1.1 The hardware hash, ZTDId and group tag

Every Autopilot device is identified by its **hardware hash** (also called the 4K hash): a blob generated by Windows from the serial number, SMBIOS, TPM and other component identifiers. Uploading the hash to the Autopilot service creates a **Windows Autopilot device** record in the tenant. That record stamps the matching Entra ID device object with a **ZTDId** (zero-touch deployment ID), stored in the device's `devicePhysicalIds` attribute as `[ZTDId]:<guid>`. An optional **group tag** is stored alongside it as `[OrderID]:<tag>`, and is how you send different kinds of device to different profiles.

Identifier	Where it lives	What it is used for
Hardware hash	CSV from the OEM, reseller or `Get-WindowsAutopilotInfo`	Proves the physical device belongs to the tenant; uploaded once per device
Serial number	Autopilot device record, Intune device record	Human lookup; matching records across portals and the asset register
ZTDId	Entra ID device object, `devicePhysicalIds`	Dynamic group membership for *all* Autopilot devices
Group tag (OrderID)	Autopilot device record, `devicePhysicalIds`	Dynamic group membership for *a kind* of device (kiosk, lab, standard)
Assigned user (optional)	Autopilot device record	Pre-fills the user name on the sign-in page; cosmetic, not security

1.2 How a device finds its profile

A registered device knows nothing until it is online. At the start of the out-of-box experience (OOBE) Windows contacts the Autopilot service with its hash; the service returns the deployment profile assigned to the device through its group. From that point the profile drives every screen the user sees, the join type, the device name and the account type.

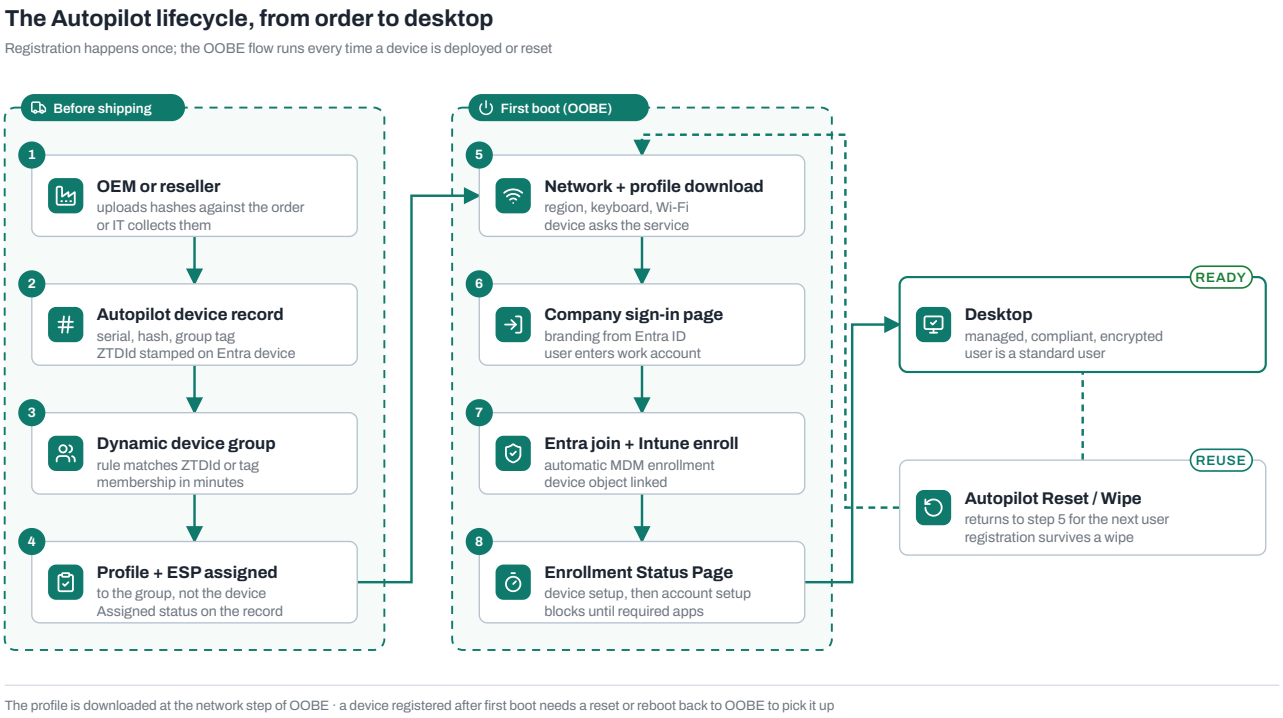


Figure 1. Registration (left) is done once per device. The OOBE flow (centre) repeats whenever the device is reset and handed to a new user.

1.3 Join types: Entra join and why not hybrid

An Autopilot profile joins the device either to **Microsoft Entra ID** only, or to on-premises Active Directory *and* Entra ID (**Microsoft Entra hybrid join**). Hybrid join requires the Intune Connector for Active Directory on a server, line of sight to a domain controller during OOBE (or a VPN that comes up before sign-in), an offline domain-join blob, and a longer, more fragile deployment. Microsoft recommends Entra join for new deployments; Autopilot device preparation supports only Entra join.

Consideration	Entra join	Hybrid join
Infrastructure	None beyond the tenant	Intune Connector for AD, OU permissions, DC reachability
Works from home on day one	Yes, any internet connection	Only with a pre-logon VPN or an on-site build
Group Policy	Not applied; use Intune Settings catalog	Applied, often duplicating Intune settings
Access to on-premises file shares	Yes, with Kerberos via Entra Kerberos / cloud trust or Hello for Business	Yes, natively
Supported by Autopilot device preparation	Yes	No
Baseline choice	Default for all new Windows devices	Documented exception only

1.4 Deployment modes

The deployment profile chooses one of three modes. They differ in who is present, how long the user waits and what hardware is required.

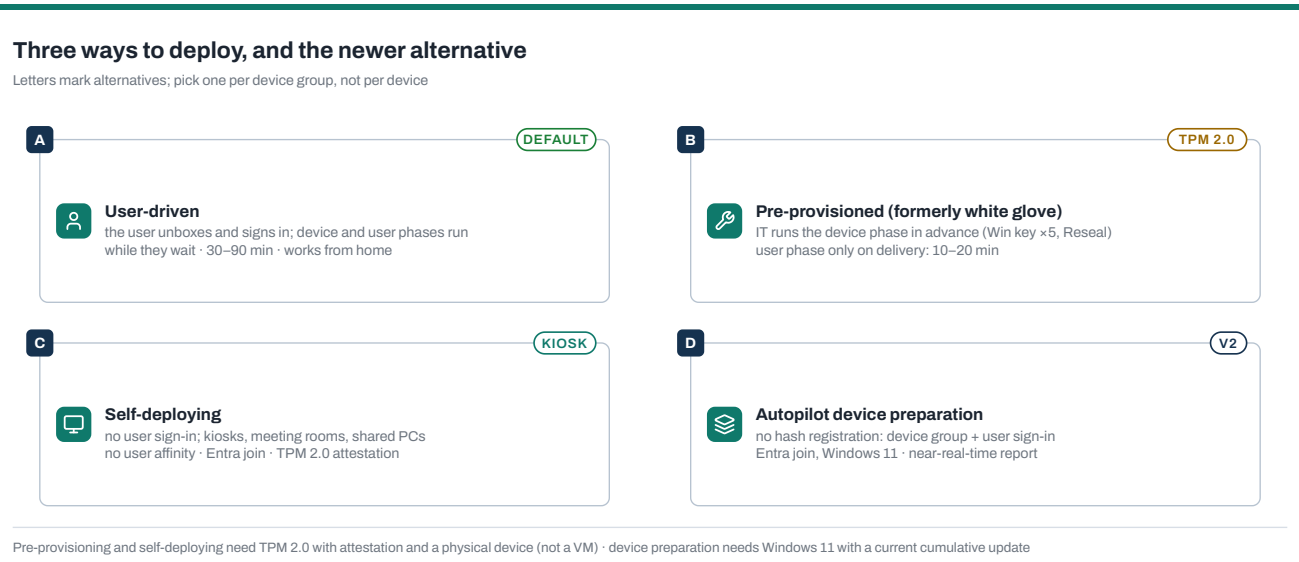


Figure 2. Modes A to C are set in a classic deployment profile. D is a separate policy type with different prerequisites.

1.5 The Enrollment Status Page

The **Enrollment Status Page (ESP)** is the progress screen that holds the user in OOBE until the device is ready. Without it, the user reaches the desktop while BitLocker, Defender, certificates and line-of-business apps are still arriving: the first hour is spent on an unprotected, half-built PC. The ESP has three phases; the first two run in the device context, the third after the user signs in.

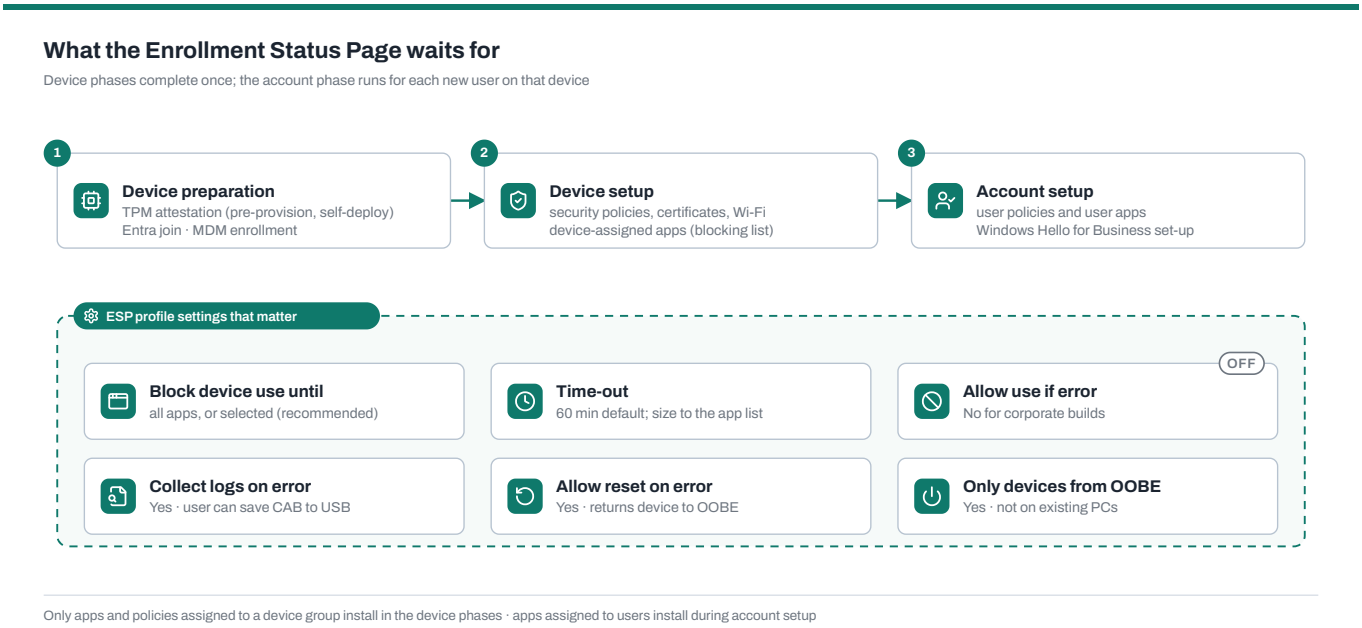


Figure 3. The three ESP phases and the profile settings this SOP uses. Each blocking app adds to the time a user waits.

1.6 Classic Autopilot versus Autopilot device preparation

Autopilot device preparation is a newer policy type that removes the hash-registration step. Instead, the device is added to a static Entra device group at enrollment time, and the policy lists the apps and scripts to install before the desktop is shown. It is simpler to start with but offers fewer OOBE controls. Where both apply to a device, a classic Autopilot profile takes precedence. This SOP builds classic Autopilot as the standard and notes the device-preparation equivalents.

Capability	Classic Autopilot (this SOP)	Autopilot device preparation
Registration	Hardware hash uploaded before first boot	None; optional corporate identifiers (serial, make, model) to restrict enrollment
Targeting	Dynamic device group (ZTDId / group tag)	Static device group owned by the *Intune Provisioning Client* service principal
Modes	User-driven, pre-provisioned, self-deploying	User-driven (plus automatic mode for Windows 365)
Join types	Entra join or hybrid join	Entra join only

Capability	Classic Autopilot (this SOP)	Autopilot device preparation
OOBE control	Hide privacy, EULA, account type, device name template	Minimal; Windows defaults for most screens
Blocking content	ESP: all or selected apps	Capped list of apps and PowerShell scripts in the policy
Reporting	Enrollment reports, ESP logs	Near-real-time Autopilot deployment report per device

CHAPTER 2

Access, Prerequisites and Planning

2.1 Level of access required

Task	Minimum role	Notes
View Autopilot devices, profiles, enrollment reports	Read Only Operator (Intune) or Global Reader	For audits and Tier 1 lookups.
Import hashes, create profiles and ESP, assign groups	Intune Administrator, or Policy and Profile Manager	Activate through PIM for the change.
Create the dynamic device group	Groups Administrator or Intune Administrator	Dynamic groups need Entra ID P1.
Automatic enrollment (MDM user scope)	Intune Administrator or Global Administrator	One-time tenant setting.
Company branding (sign-in page)	Organizational Branding Administrator	Shown at the company sign-in step of OOBE.
Collect a hash on a device (`-Online`)	Intune Administrator on the Graph connection; local admin on the device	Or export to CSV and have an admin import it.
Autopilot Reset, wipe, delete record	Help Desk Operator (reset) · Intune Administrator (delete)	Wipe may require multi-admin approval (SOP-EM-003).

2.2 Licensing and platform

- Each user needs **Microsoft Intune** and **Microsoft Entra ID P1** (both included in Microsoft 365 E3/E5, Business Premium and EMS E3/E5).
- Windows 10/11 **Pro, Business, Enterprise or Education**, a currently supported release. Home editions cannot join Entra ID.
- Pre-provisioning and self-deploying need **TPM 2.0** with device health attestation, and a physical device.
- Autopilot device preparation needs Windows 11 with a current cumulative update; check Microsoft's minimum build before relying on it.

2.3 Prerequisites checklist

- ☐ Automatic enrollment set: Devices › Enrollment › Windows › Automatic Enrollment › **MDM user scope** *All* (or a group that contains every Autopilot user); MAM user scope *None* unless deliberately used.
- ☐ Entra ID › Devices › Device settings: *Users may join devices to Microsoft Entra* = All (or the Autopilot user group); device limit per user high enough (default 50).

- [ ] Windows enrollment restrictions allow Windows (MDM); personally owned Windows devices may be blocked, Autopilot devices are corporate by definition.
- [ ] Company branding configured (logo, sign-in page text, tenant name) so users recognise the sign-in page.
- [ ] Network allows the Autopilot, Entra ID, Intune, Windows Update, Delivery Optimization and TPM attestation endpoints (no TLS inspection on them).
- [ ] Required apps packaged and assigned to a **device** group (Company Portal, Defender configuration, VPN client, line-of-business apps). Keep the blocking list short.
- [ ] Compliance, BitLocker, Windows Hello for Business, update ring (SOP-EM-005) and security baseline policies assigned to the same device group.
- [ ] Procurement agreement with the OEM or reseller to register hashes against every order, with the tenant ID and the group tag to use.

2.4 Risk and planning notes

Action	Risk	Notes
Read Autopilot records and reports	READ-ONLY	Always first when troubleshooting.
Import hashes; create groups, profiles, ESP	LOW	Affects only devices that are not yet deployed.
Assign a profile to an existing fleet with *Convert all targeted devices*	MODERATE	Registers every Entra-joined device in the group; they get the profile on their next reset.
Change the ESP blocking-app list	MODERATE	A broken app now blocks every new deployment. Test on one device.
Autopilot Reset a user's device	MODERATE	Removes user data and apps; keeps enrollment. Confirm OneDrive backup.
Wipe, delete Intune record, deregister	HIGH	Irreversible. Deregistering removes the device from the tenant; it will not find its profile again.

2.5 Decisions to make up front

Decision	Options	Baseline choice
Join type	Entra join; hybrid join	Entra join
Mode	User-driven; pre-provisioned; self-deploying	User-driven for staff; pre-provisioned for executives and lab; self-deploying for kiosks
Group tags	None; one per persona; one per site	One per persona: `STD`, `EXEC`, `LAB`, `KIOSK`
Device name template	OEM default; `%SERIAL%`; `%RAND:x%`	`CTS-%SERIAL%` (15 characters maximum, truncated serial)

Decision	Options	Baseline choice
User account type	Administrator; Standard	Standard; elevation through LAPS or Endpoint Privilege Management
ESP blocking apps	All; selected; none	Selected: Company Portal, Defender onboarding, VPN, one core LOB app

CHAPTER 3

Registering Devices

A device must be registered before its first boot reaches the network step of OOBE. There are three ways in; all of them end in the same Autopilot device record.

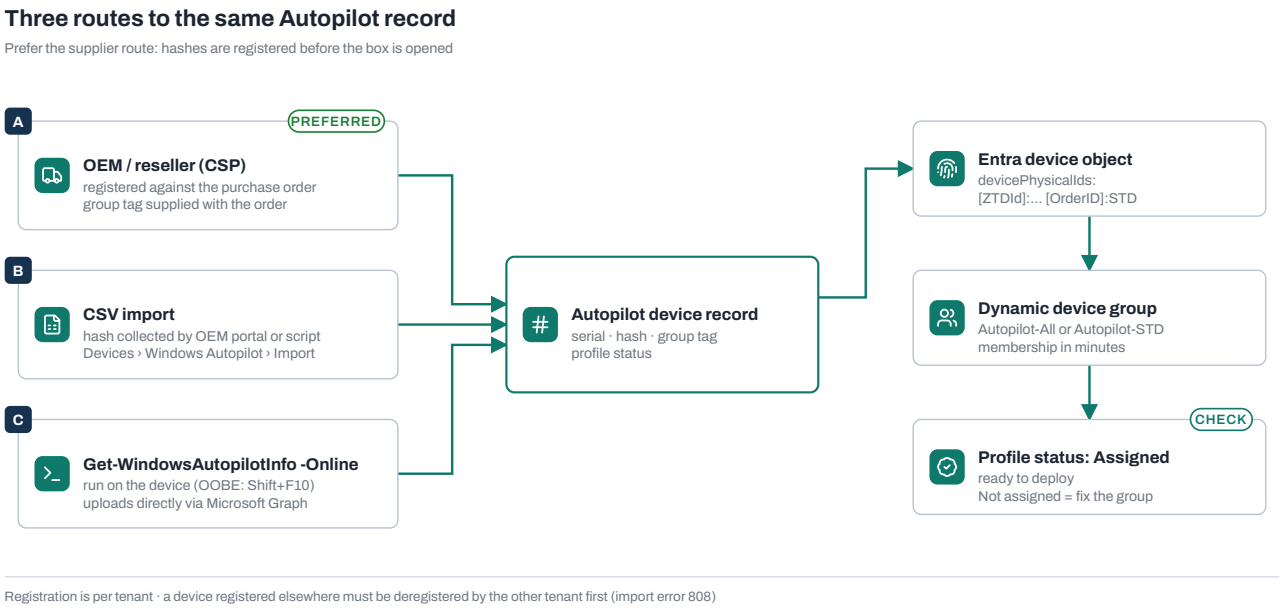


Figure 4. All three routes create the same record. The ZTDId on the Entra device object is what puts the device in the group.

3.1 Collecting a hash from a device

For devices already in hand (pilot devices, stock without supplier registration), collect the hash with Microsoft's `Get-WindowsAutopilotInfo` script from the PowerShell Gallery. On a new device, press Shift+F10 at the first OOBE screen to open a command prompt, then start PowerShell.

ON THE DEVICE · UPLOAD THE HASH DIRECTLY — POWERSHELL

```powershell
# Run in an elevated PowerShell (at OOBE: Shift+F10, then type powershell)
Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned -Force
Install-Script -Name Get-WindowsAutopilotInfo -Force
# -Online signs in to Microsoft Graph interactively as an Intune Administrator
# -GroupTag sets [OrderID]; -Assign waits until a profile is assigned
Get-WindowsAutopilotInfo -Online -GroupTag "STD" -Assign
```

ON THE DEVICE · EXPORT TO CSV FOR AN ADMIN TO IMPORT — POWERSHELL

```powershell
New-Item -Type Directory -Path C:\HWID -Force | Out-Null
Get-WindowsAutopilotInfo -OutputFile C:\HWID\$env:COMPUTERNAME.csv
# CSV columns: Device Serial Number, Windows Product ID, Hardware Hash, (Group Tag), (Assigned User)
```

WATCH OUT · THE CSV FORMAT IS STRICT

Import fails if the file is re-saved by Excel (which can change encoding and quoting), has extra columns, or has more than 500 rows. Keep the header exactly as exported, ANSI or UTF-8 without extra BOM characters, and split large files.

STEP 1 **Import a CSV in the Intune admin center**

1. Go to `Devices` › `Enrollment` › `Windows` › `Devices` (under *Windows Autopilot*).
2. Select `Import`, choose the CSV and select `Import` again. Import runs in the background and can take 15 minutes or more.
3. Select `Sync` and `Refresh`. Each serial appears with *Profile status: Not assigned* until the group and profile are in place.
4. Open a device record to confirm the **Group tag** and, if used, **User** fields. Correct a tag here rather than re-importing.

3.2 Target devices with a dynamic group

Profiles are assigned to Entra device groups. Use dynamic membership rules so every newly registered device lands in the right group without manual work.

Group	Membership type	Rule
Autopilot-All	Dynamic Device	`(device.devicePhysicalIDs -any (_ -startsWith "[ZTDid]"))`
Autopilot-STD	Dynamic Device	`(device.devicePhysicalIds -any (_ -eq "[OrderID]:STD"))`
Autopilot-EXEC	Dynamic Device	`(device.devicePhysicalIds -any (_ -eq "[OrderID]:EXEC"))`
Autopilot-KIOSK	Dynamic Device	`(device.devicePhysicalIds -any (_ -eq "[OrderID]:KIOSK"))`

STEP 2 **Create the dynamic group**

1. Entra admin center › `Groups` › `All groups` › `New group`. Group type *Security*, name `Autopilot-STD`.
2. `Membership type` *Dynamic Device* › `Add dynamic query` › `Edit` the rule syntax and paste the rule from the table.
3. Use `Validate Rules` with one known device to confirm it evaluates to *true*, then `Save` and `Create`.
4. Wait for membership processing (usually minutes; longer in large tenants) before assigning the profile.

CHAPTER 4

Building the Profile and the Enrollment Status Page

STEP 1 Create the deployment profile

1. Devices › Enrollment › Windows › Deployment profiles › Create profile › *Windows PC*.
2. Name `AP-UserDriven-STD`. **Convert all targeted devices to Autopilot**: *No* for the standard profile (see the note below).
3. Out-of-box experience: Deployment mode *User-Driven*; Join to Microsoft Entra ID as *Microsoft Entra joined*.
4. Microsoft Software License Terms *Hide*; Privacy settings *Hide*; Hide change account options *Hide*; User account type *Standard*.
5. Allow pre-provisioned deployment *Yes*; Language (Region) *Operating system default*; Automatically configure keyboard *Yes*.
6. Apply device name template *Yes* › `CTS-%SERIAL%`. Names longer than 15 characters are truncated.
7. Assignments › Include `Autopilot-STD` (and exclude nothing). Create.

Convert all targeted devices registers existing Entra-joined devices in the group as Autopilot devices. Use it deliberately, in its own profile, when bringing an existing fleet under Autopilot.

STEP 2 Create the Enrollment Status Page profile

1. Devices › Enrollment › Windows › Enrollment Status Page › Create. Name `ESP-STD`.
2. Show app and profile configuration progress *Yes*; Show an error when installation takes longer than *60* minutes.
3. Show custom message *Yes*: "Setup is still running. If it fails, contact the service desk on ext. 4357 and quote the error code."
4. Turn on log collection and diagnostics page for end users *Yes*; Only show page to devices provisioned by OOBE *Yes*.
5. Block device use until all apps and profiles are installed *Yes*; Allow users to reset device if installation error occurs *Yes*; Allow users to use device if installation error occurs *No*.
6. Block device use until these required apps are installed › *Selected*: Company Portal, Microsoft Defender configuration, VPN client, core LOB app.
7. Assign to `Autopilot-STD`. If more than one ESP profile targets a device, the one with the highest priority wins; set the order in the ESP list.

GOOD PRACTICE · KEEP THE BLOCKING LIST SHORT AND RELIABLE

Every blocking app is a single point of failure for every new deployment. Block only on what the user needs on day one and what protects the device; let everything else install after the desktop appears. Win32 apps and Microsoft Store apps should not mix with legacy MSI line-of-business apps in the device phase, because Win32 and MSI installs can collide during ESP.

4.1 Policies that must land during the device phase

Policy	Assigned to	Why it belongs in the ESP
BitLocker (Endpoint security › Disk encryption)	Autopilot-All	Encryption starts silently before user data exists; key escrowed to Entra ID.
Microsoft Defender for Endpoint onboarding	Autopilot-All	Device is protected and reporting before first sign-in.
Compliance policy (Windows)	Autopilot-All	Feeds Conditional Access CA006 (SOP-ID-001); device must be compliant to reach Exchange and SharePoint.
Windows Hello for Business	Users or devices	Provisioned at first sign-in, so the user leaves with a phishing-resistant credential.
Update ring and feature update policy	Autopilot-All	Device patches from day one (SOP-EM-005).
Windows LAPS	Autopilot-All	Local administrator password rotated and escrowed; no shared local admin.

CHAPTER 5

Deploying: the OOBE Experience

5.1 User-driven deployment

STEP 1 **What the user does (and what is happening)**

1. Power on, choose region and keyboard, connect to Wi-Fi or Ethernet. *The device contacts the Autopilot service and downloads its profile.*
2. The **Welcome to Contoso** sign-in page appears with company branding. The user enters their work email and password and completes MFA. *Conditional Access applies here.*
3. The device joins Entra ID and enrolls in Intune. *Device preparation and device setup phases run; the ESP shows progress.*
4. After a restart (if required) the user signs in again; *account setup* runs, user apps and policies install.
5. Windows Hello for Business set-up prompts for a PIN and biometrics. The desktop appears; OneDrive Known Folder Move starts.

Typical time: 30–90 minutes, most of it in device setup. Pre-provisioning moves that part to IT.

5.2 Pre-provisioning (technician phase)

For executives, lab instruments and remote starters, IT or the reseller runs the device phases before shipping. The user then only waits for account setup.

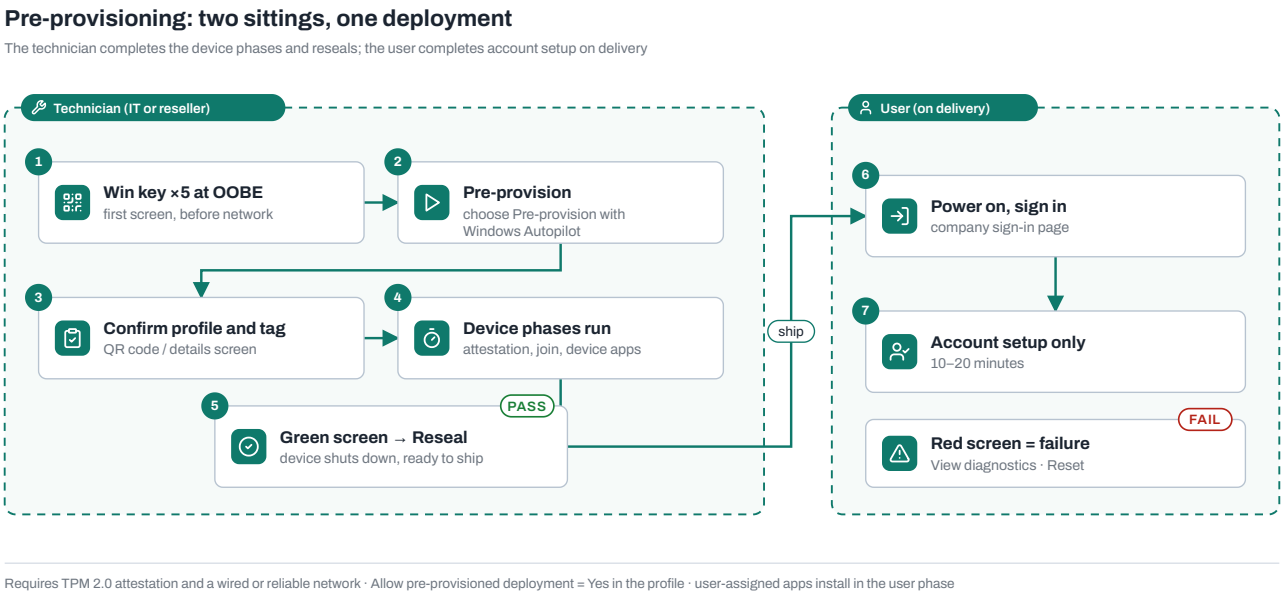


Figure 5. A pre-provisioned device leaves IT already joined, enrolled and carrying its device apps; the user finishes in minutes.

5.3 Self-deploying (kiosk and shared devices)

Assign a profile with **Deployment mode: Self-Deploying** to `Autopilot-KIOSK`. The device needs no user: it attests its TPM, joins Entra ID, enrolls and applies the kiosk or shared-PC configuration on its own. Because there is no user, user-assigned apps and user policies do not apply; target everything at the device group, and pair it with a kiosk (Assigned Access) or shared multi-user device profile.

CHAPTER 6

Method B · PowerShell and Microsoft Graph

Use Graph for bulk work: importing hashes from a supplier file, reporting on profile status across the fleet, and checking a single device during a support call.

6.1 Connect

CONNECT · DELEGATED, LEAST PRIVILEGE — POWERSHELL

```
Install-Module Microsoft.Graph -Scope CurrentUser        # once
Connect-MgGraph -Scopes
"DeviceManagementServiceConfig.ReadWrite.All","DeviceManagementManagedDevices.Read.All","Group.Read.All" -
NoWelcome
```

6.2 Import hashes from a CSV

BULK IMPORT · GRAPH REST VIA INVOKE-MGGRAPHREQUEST — POWERSHELL

```
$rows = Import-Csv C:\HWID\order-4711.csv
foreach ($r in $rows) {
  $body = @{
    serialNumber       = $r.'Device Serial Number'
    productKey         = $r.'Windows Product ID'
    hardwareIdentifier = $r.'Hardware Hash'
    groupTag           = "STD"
  } | ConvertTo-Json
  Invoke-MgGraphRequest -Method POST -Body $body -ContentType "application/json" `
    -Uri "https://graph.microsoft.com/v1.0/deviceManagement/importedWindowsAutopilotDeviceIdentities"
}
# Ask the service to process the imports, then check state
Invoke-MgGraphRequest -Method POST -Uri
"https://graph.microsoft.com/v1.0/deviceManagement/windowsAutopilotSettings/sync"
```

6.3 Report profile status across the fleet

WHICH REGISTERED DEVICES ARE NOT READY TO DEPLOY? — POWERSHELL

```
$uri = "https://graph.microsoft.com/v1.0/deviceManagement/windowsAutopilotDeviceIdentities"
$all = @(); do { $page = Invoke-MgGraphRequest -Uri $uri; $all += $page.value; $uri = $page.'@odata.nextLink' }
while ($uri)
$all | Where-Object { $_.deploymentProfileAssignmentStatus -notmatch 'assigned' } |
  Select-Object serialNumber, groupTag, model, deploymentProfileAssignmentStatus, lastContactedDateTime |
  Sort-Object groupTag | Format-Table -AutoSize
```

> **NOTE · COMMUNITY MODULE**
>
> The `WindowsAutopilotIntune` module from the PowerShell Gallery wraps the same endpoints (`Get-AutopilotDevice` , `Add-AutopilotImportedDevice` , `Invoke-AutopilotSync`). It is community-maintained; pin the version you test and review it before use in a regulated environment.

CHAPTER 7

Verification, Reset and Troubleshooting

7.1 Verification checklist

Check	Where	Expected
Device registered and tagged	Windows Autopilot devices	Serial present; group tag correct; Profile status *Assigned*
Device in the right group	Entra ID › device › Groups	Member of `Autopilot-All` and its persona group
Deployment completed	Devices › Monitor › Autopilot deployments	Status *Success*; deployment and ESP durations recorded
Join and enrollment	Device: `dsregcmd /status`	*AzureAdJoined : YES*; MDM URL present
Encryption and protection	Intune device › Device configuration / Defender	BitLocker on; Defender onboarded; compliant
User experience	Test sign-in	Standard user; Windows Hello set up; Company Portal shows required apps installed
Name	Intune device record	`CTS-<serial>`

7.2 Reset, wipe and deregister

Action	What happens	When to use
Autopilot Reset (remote)	Removes user files, apps and settings; keeps Entra join, Intune enrollment and Wi-Fi; returns to sign-in	Handing a device to a new user in the same role
Wipe	Factory reset; device returns to OOBE and re-runs Autopilot (registration survives)	Device rebuild, malware, persona change
Retire	Removes company data and management; personal data kept	Not used for corporate Autopilot devices
Delete Intune record, then deregister	Removes the Autopilot registration; device will no longer find its profile	Disposal, sale, return to OEM, move to another tenant

WARNING · DEREGISTER IN THE RIGHT ORDER

Delete the device from Intune (`Devices` › `Windows` › device › `Delete`) before deleting the Autopilot record; a device still enrolled cannot be deregistered. Record the serial and the reason in the asset register. Wipes on high-value devices may require a second administrator's approval (SOP-EM-003).

7.3 Troubleshooting

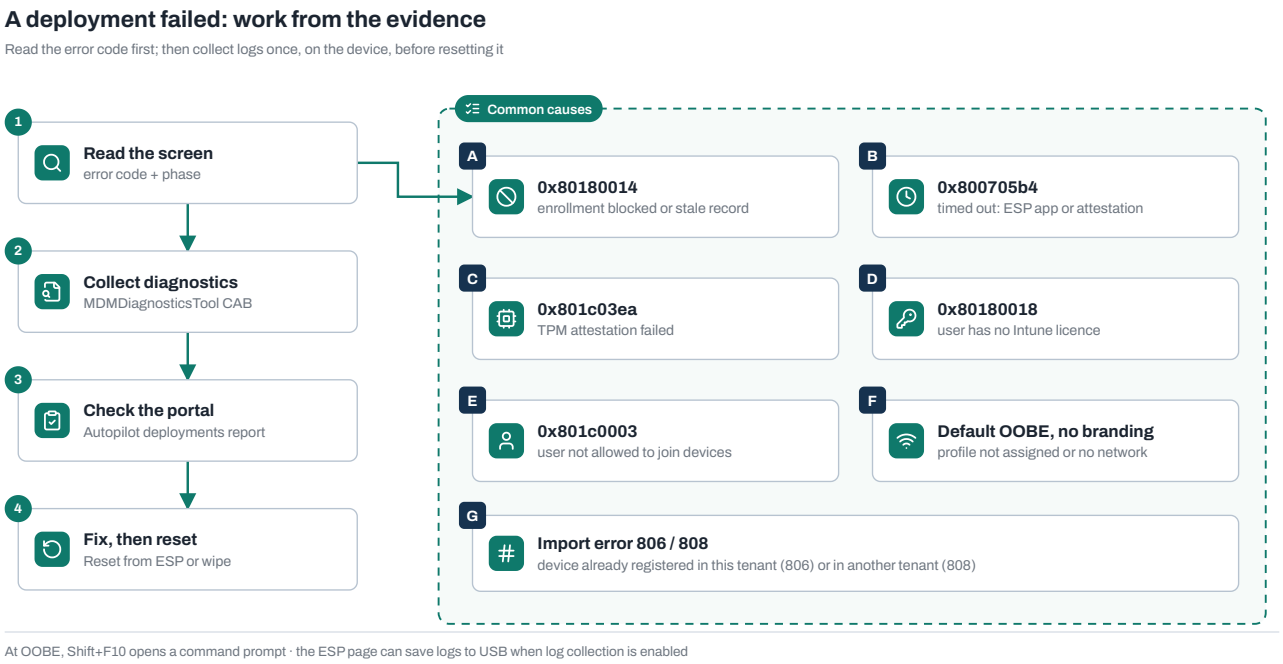


Figure 6. Diagnosis starts from the code on the screen. Collect the CAB before resetting, because a reset discards the evidence.

ON THE FAILED DEVICE · COLLECT EVERYTHING ONCE — POWERSHELL

```
# At the ESP error page: Shift+F10 opens a command prompt
mkdir C:\Diag
MdmDiagnosticsTool.exe -area "Autopilot;DeviceEnrollment;DeviceProvisioning;TPM" -cab C:\Diag\autopilot.cab
# Quick views
dsregcmd /status
Get-WinEvent -LogName "Microsoft-Windows-ModernDeployment-Diagnostics-Provider/Autopilot" -MaxEvents 50 |
  Select-Object TimeCreated, Id, Message | Format-List
Get-WinEvent -LogName "Microsoft-Windows-DeviceManagement-Enterprise-Diagnostics-Provider/Admin" -MaxEvents 50
|
  Where-Object LevelDisplayName -eq "Error" | Format-List TimeCreated, Id, Message
```

Symptom	Likely cause	Fix
OOBE shows the consumer setup, no company page	Profile not assigned, device not in group, or no network at the profile step	Check Profile status *Assigned*; fix the group rule; then return the device to OOBE with a wipe and retry.

Symptom	Likely cause	Fix
0x80180014 at enrollment	Windows MDM blocked by enrollment restrictions, or an old Intune record for the device	Allow Windows (MDM) for the user; delete the stale Intune device record and retry.
ESP times out in device setup (0x800705b4)	A blocking app failed or is too slow; Win32/MSI install collision	Find the app in the ESP details and IME log (`C:\ProgramData\Microsoft\IntuneManagementExtension\Logs`); fix or remove from the blocking list; raise the time-out only after.
Pre-provisioning red screen at attestation (0x801c03ea)	TPM not 2.0, firmware TPM issue, or attestation endpoints blocked	Update BIOS/TPM firmware; clear TPM; confirm network allows attestation; use user-driven instead.
0x80180018	User lacks an Intune licence	Assign the licence group; wait for sync; retry.
Device name not applied	Template invalid or > 15 characters, or hybrid join (template not supported)	Use `CTS-%SERIAL%` ; set naming in the domain-join profile for hybrid.

CHAPTER 8

Summary, Glossary and Review

8.1 Chapter summary

- Autopilot transforms the OEM installation of Windows into a managed device; nothing is imaged.
- The hardware hash registers the device; the ZTDId and group tag put it in a dynamic group; the group carries the profile and ESP.
- Entra join is the standard. Hybrid join is a documented exception; device preparation supports Entra join only.
- User-driven for staff, pre-provisioned for executives and labs, self-deploying for kiosks; each has its own profile and group.
- The ESP blocks the desktop until security policies and a short list of essential apps are in place.
- Collect diagnostics before resetting; delete the Intune record before deregistering.

Check your understanding

1. What links an uploaded hardware hash to the dynamic group that assigns the deployment profile?
2. A device was registered after it had already been switched on and set up. What must happen before it picks up the profile?
3. Why should the ESP block on selected apps rather than all apps?
4. Which two hardware or platform requirements do pre-provisioning and self-deploying share?
5. A device shows import error 808. What does it mean and who must act?

8.2 Glossary

Hardware hash	Device identifier generated by Windows from hardware components; registers the device with Autopilot.
ZTDId	Zero-touch deployment ID written to the Entra device object's devicePhysicalIds when a device is registered.
Group tag	Optional OrderID value on the Autopilot record used to target device personas.
Deployment profile	Autopilot settings for mode, join type, OOBE screens, account type and device name.
Enrollment Status Page	OOBE progress page that blocks use until required policies and apps are installed.
Pre-provisioning	Running the device phases in advance (Win key ×5) and resealing; formerly white glove.
Self-deploying	Deployment without a user, for kiosks and shared devices; requires TPM 2.0 attestation.

Autopilot device preparation	Newer policy type using a device group instead of hash registration; Entra join only.
Autopilot Reset	Returns a device to a business-ready state for a new user while keeping enrollment.

8.3 Framework references

Framework	Controls addressed
NIST SP 800-53 Rev. 5	CM-2 Baseline Configuration · CM-6 Configuration Settings · CM-8 System Component Inventory · IA-3 Device Identification and Authentication · SC-28 Protection of Information at Rest (BitLocker)
CIS Controls v8	1.1 Enterprise asset inventory · 4.1 Secure configuration process · 4.7 Manage default accounts
FDA 21 CFR Part 11 / GAMP 5	Documented, repeatable build of computerized-system clients; configuration evidence retained per device